

Individual Threestudio Setup & Debugging Guide for local PC

DISCLAIMERS:

- This setup is made for you to test that threestudio docker setup runs by itself on your PC before running the combined threestudio with FoundationPose setup
- It is assumed the user has already installed docker and set it with admin permissions

Step 0: Setup docker to run without sudo

```
Unset
sudo groupadd docker
sudo usermod -aG docker $USER
newgrp docker
docker ps
```

Step 1: Clone The Repository (if you haven't done so already)

```
Unset
https://github.com/JustinLungu/FoundationPose-BachelorThesis.git
cd FoundationPose-BachelorThesis
```

Step 2: Setup the virtual environment

First make sure you have all the required dependencies to run all pipelines in the project, including the population of the repository:

```
Unset
cd ..
conda env create -f environment.yml
#OR
pip install -r requirements.txt
```

Step 3: Populate the project completely

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```
python populate_pipelines.py
```

To download all the necessary data, including the threestudio setup environment, such that all pipelines can be run successfully.

Step 4: Build the Threestudio docker

If you did not do it already from the FoundationPose setup, populate the repository with all the extra data:

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```
cd threestudio/docker  
docker build -t threestudio_custom threestudio
```

Step 4: Run the Threestudio docker

If you did not do it already from the FoundationPose setup, populate the repository with all the extra data:

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```
./run_container.sh
```

Step 4: Setup Huggingface Token

Note: you need to have a huggingface account and create an access token which you will use in this step:

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```
huggingface-cli login  
#then ctrl+shift+v the token
```

Step 5: Test threestudio with the simplest project

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```
python launch.py --config configs/dreamfusion-sd.yaml --train --gpu 0  
system.prompt_processor.prompt="a zoomed out DSLR photo of a baby bunny  
sitting on top of a stack of pancakes"
```

Note: if it's the first time you run a specific model, it will take some time to download the necessary files and weights for the model to run.